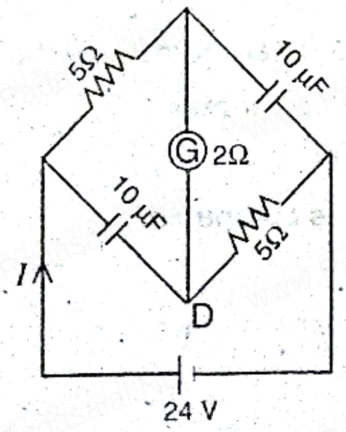
Standard 12

Time Allowed: 3.00 Hours **PHYSICS** Maximum Marks: 70

Part - I

I. Choose the correct answer: 15 x 1 = 15

1. A parallel plate capacitor stores at charge 'Q' at a voltage 'V'. Suppose the area of the parallel plate capacitor and the distance between the plates are each doubled, then which of the following quantity will change?
   a) capacitance b) charge
   c) voltage d) energy density
2. The work done in moving a charge of 100μc along the equatorial line of an electric dipole between two points which are 1m from each other is
   a) 9×10^{-3} J b) 9×10^{3} J
   c) 80 J d) zero
3. Which of the following electromagnetic radiations is used for viewing objects through fog?
   a) micro wave b) gamma rays
   c) X-rays d) infrared
4. The electric and magnetic fields of an electromagnetic wave are
   a) in phase and the perpendicular to each other
   b) out of phase and not perpendicular to each other
   c) in phase and not perpendicular to each other
   d) out of phase and perpendicular to each other
5. The frequency range of infrared radiation is
   a) 10^{11} Hz to 4×10^{14} Hz a) 4×10^{14} Hz to 8×10^{14} Hz
   c) 10^{9} Hz to 10^{11} Hz d) 8×10^{14} Hz to 10^{17} Hz
6. When steady current flows the circuit, what is the current through the galvanometer in the circuit given below?



a) 1 A b) 1.5 A
c) 2 A d) 2.5 A

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7. In India electricity is supplied for domestic use at 220 V and it is supplied at 110V in USA. If the resistance of 60W bulb for use in India is 'R', the resistance of a 60W bulb for use in USA will be

a) R b) 2R c) $R/4$ d) $R/2$

8. In Joule's heating law, when 'R' and 't' are constants, if 'H' is taken along y-axis, and I^2 along x-axis, The graph is

a) straight line b) parabola
c) circle d) ellipse

9. A circular coil of radius 5cm and 50 turns carries a current of 3A. The magnetic dipole moment of the coil is nearly

a) 1 Am^2 b) 1.2 Am^2
c) 0.5 Am^2 d) 0.8 Am^2

10. The vertical component of Earth's magnetic field at a place is equal to $\sqrt{3}$ times the horizontal component. What is the value of angle of dip at this place?

a) 30° b) 60° c) 90° d) 120°

11. When the current changes from +2A to −2A in 0.05s, an emf of 8V is induced in a coil. The co-efficient of self-induction of the coil is

a) 0.2 H b) 0.4 H c) 0.8 H d) 0.1 H

12. A step down transformer reduces the supply voltage from 220V to 11V and increases the current from 6A to 100A. Then its efficiency is

a) 1.2 b) 0.83 c) 0.12 d) 0.9

13. In a series resonant RLC circuit, the voltage across 100 Ω resistor is 40V. The resonant frequency 'ω' is 250 rad/s. If the value of 'C' is 4μF, then the voltage across 'L' is

a) 600 V b) 4000 V c) 400 V d) 1 V

14. The speed of light in an isotropic medium depends on

a) its intensity b) its wavelength c) nature of propagation d) all

15. Stars twinkle due to _____ of light.

a) reflection b) total internal reflection
c) refraction d) polarization

Part - II

II. Answer any 6 questions. (Q.No.24 is compulsory) 6 x 2 = 12

16. Define Electric dipole moment
17. Mention any two properties of equipotential surface.
18. State Kirchoff's voltage rule.

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19. Define One Ampere (on the basis of magnetic force)
20. State Ampere's circuital law.
21. Define Q factor of RLC series circuit.
22. What is displacement current?
23. An object is placed at a distance of 20 cm from a concave mirror of focal length 15 cm. Calculate the image distance.
24. The resistance of a nichrome wire at 20°C is 10 Ω. If its temperature co-efficient of resistivity of nichrome is 0.004/°C, find the resistance of the wire at boiling point of water. Comment on the result.

Part - III

III. Answer any 6 questions. (Q.No.33 is compulsory) 6 x 3 = 18

25. Mention any three differences between polar molecules and non-polar molecules.
26. What is Seebeck effect? Mention any two uses of Seebeck effect.
27. The repulsive force between two magnetic poles in air is 9×10^{-3} N. If the two poles are equal in strength and are separated by a distance of 10 cm, calculate pole strength of each pole.
28. Give an account of magnetic Lorentz force.
29. Obtain the expression for induced emf by changing the area enclosed by the coil.
30. Write a note on ultraviolet rays.
31. Obtain the expression for energy stored in a capacitor.
32. Derive the relation between radius of curvature and focal length of a concave mirror.
33. Find the impedance of a series RLC circuit, if the inductive reactance, capacitive reactance and resistance are 184 Ω, 144Ω and 30Ω respectively. Also calculate the phase angle between voltage and current.

Part - IV

IV. Answer all the questions. 5 x 5 = 25

34. a) State Gauss law. Obtain the expression for electric field due to an charged infinite plane sheet.

(OR)

b) Explain The principle, construction and working of a transformer.

35. a) Explain the types of emission spectrum.

(OR)

b) Obtain the expression for magnetic field at a point on the axial line of a bar magnet.

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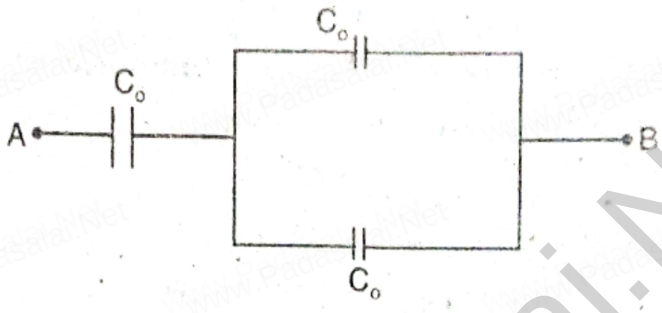
36. a) i) List out any six properties of electromagnetic waves.

ii) Calculate the speed of the electromagnetic wave in a medium if the amplitude of electric and magnetic fields are 3×10^4 NC^{-1} and 2×10^{-4}T respectively.

(OR)

b) i) Obtain the expression for effective capacitance when three capacitors of capacitance C_1, C_2 and C_3 are connected in series.

ii) Find the effective capacitance between 'A' and 'B' in the circuit shown below.



37. a) Describe Fizeau's method to determine the velocity of light in air.

(OR)

b) Obtain the expression for magnetic force acting on a current carrying conductor placed in a uniform magnetic field.

38. a) Describe the potentiometer experiment to compare the emfs of two cells.

(OR)

b) Find out the phase relationship between voltage and current in a pure inductive a.c. and current in a pure circuit. Draw the phasor diagram.

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